

Unit 1: Kotlin Basics

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```
fun main() {  
    // 1. Promotional Sale Program  
    println("=== Promotional Sale ===")  
    var discountPercentage: Int = 0  
    var offer: String = ""  
    val item = "Google Chromecast"  
    discountPercentage = 20  
    offer = "Sale - Up to $discountPercentage% discount on $item! Hurry up!"  
    println(offer)  
    println()  
}
```

```
// 2. Party Size Program  
println("=== Party Size ===")  
val numberOfAdults = 20  
val numberOfKids = 30  
val total = numberOfAdults + numberOfKids  
println("The total party size is: $total")  
println()
```

```
// 3. Salary Program  
println("=== Salary ===")  
val baseSalary = 5000  
val bonusAmount = 1000  
val totalSalary = baseSalary + bonusAmount  
println("Congratulations for your bonus! You will receive a total of $totalSalary (additional bonus).")  
println()
```

```
// 4. Basic Math Operations
println("=== Math Operations ===")
val firstNumber = 10
val secondNumber = 5
val thirdNumber = 8

val result = add(firstNumber, secondNumber)
val anotherResult = add(firstNumber, thirdNumber)
val subtractionResult = subtract(firstNumber, secondNumber)

println("$firstNumber + $secondNumber = $result")
println("$firstNumber + $thirdNumber = $anotherResult")
println("$firstNumber - $secondNumber = $subtractionResult")
println()
```

```
// 5. Gmail Alert Message
println("=== Gmail Alerts ===")
val firstUserEmailId = "user_one@gmail.com"
println(displayAlertMessage(emailId = firstUserEmailId))
println()

val secondUserOperatingSystem = "Windows"
val secondUserEmailId = "user_two@gmail.com"
println(displayAlertMessage(secondUserOperatingSystem, secondUserEmailId))
println()

val thirdUserOperatingSystem = "Mac OS"
val thirdUserEmailId = "user_three@gmail.com"
println(displayAlertMessage(thirdUserOperatingSystem, thirdUserEmailId))
println()
```

```

// 6. Pedometer Program
println("=== Pedometer ===")
val steps = 4000
val caloriesBurned = stepsToCalories(steps)
println("Walking $steps steps burns $caloriesBurned calories")
println()

// 7. Compare Two Numbers (Screen Time)
println("=== Screen Time Comparison ===")
val todayMinutes = 120
val yesterdayMinutes = 150
val isTodayMore = compareScreenTime(todayMinutes, yesterdayMinutes)
println("Did I spend more time today than yesterday? $isTodayMore")
}

```

```

// Reusable functions
fun add(a: Int, b: Int): Int {
    return a + b
}

fun subtract(a: Int, b: Int): Int {
    return a - b
}

fun displayAlertMessage(operatingSystem: String = "Unknown OS", emailId: String): String {
    return "There's a new sign-in request on $operatingSystem for your Google Account $emailId."
}

```

```

fun stepsToCalories(numberOfSteps: Int): Double {
    val caloriesPerStep = 0.04
    val totalCaloriesBurned = numberOfSteps * caloriesPerStep
    return totalCaloriesBurned
}

fun compareScreenTime(todayMinutes: Int, yesterdayMinutes: Int): Boolean {
    return todayMinutes > yesterdayMinutes
}

```

Code Output:

```

=== Promotional Sale ===
Sale - Up to 20% discount on Google Chromecast! Hurry up!

=== Party Size ===
The total party size is: 50

=== Salary ===
Congratulations for your bonus! You will receive a total of 6000 (additional bonus).

```

```
=== Math Operations ===
```

```
10 + 5 = 15
```

```
10 + 8 = 18
```

```
10 - 5 = 5
```

```
=== Gmail Alerts ===
```

```
There's a new sign-in request on Unknown OS for your Google Account user_one@gmail.com.
```

```
There's a new sign-in request on Windows for your Google Account user_two@gmail.com.
```

```
There's a new sign-in request on Mac OS for your Google Account user_three@gmail.com.
```

```
=== Pedometer ===
```

```
Walking 4000 steps burns 160.0 calories
```

```
=== Screen Time Comparison ===
```

```
Did I spend more time today than yesterday? false
```